

# Unit 1

## Matrices

$$A = [a_{ij}]_{m \times n}$$

$i \rightarrow i^{\text{th}}$  row

$m \rightarrow$  number of rows

$j \rightarrow j^{\text{th}}$  Column

$n \rightarrow$  Column,

Types  $\rightarrow$  (i) Row matrix  $\Rightarrow [a \ b \ c]$

(ii) Column matrix  $\Rightarrow \begin{bmatrix} a \\ b \end{bmatrix}$

(iii) Null Matrix (zero)  $\Rightarrow [0]$

(iv) Square  $\Rightarrow m = n$

(v) Diagonal  $\Rightarrow$  main diagonal element  $\neq 0$

$$\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

(vi) Scalar matrix  $\Rightarrow$  diagonal element constant

(vii) Unit matrix  $\Rightarrow I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$   
(Identity)

(viii) Transpose  $\Rightarrow m \leq n \Rightarrow A^T$

(ix) Symmetric  $\Rightarrow A = A^T$

(x) Skew symmetric  $\Rightarrow A = -A^T$

(xi) Trace of matrix  $\Rightarrow$  sum of diagonal element  
 $= \text{tr}(A)$

(xii) Triangular  $\rightarrow$  Upper  $\Rightarrow \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$   
 $\rightarrow$  Lower  $\Rightarrow \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix}$

(xiii) Equal matrix

$\Rightarrow$  order same

corresponding element are equal.

(xiv) Orthogonal  $\Rightarrow A A^T = A^T A = I$

(xv) Idempotent  $\Rightarrow A^2 = A$

(xvi) Involuntary  $\Rightarrow A^2 = I$

Date: / /

Algebra (i) Addition (Same order)

(ii) Subtraction

(iii) Multiplication

(a) Scalar  $\times$  matrix  $\Rightarrow kA$

(b) two matrix  $\Rightarrow A, B$  where  $A = [a_{ij}]_{m \times n}$

$C = AB$  where  $B = [b_{jk}]_{n \times p}$   
where  $C = [c]_{m \times p}$

## Determinant

order 1  $\Rightarrow \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \Rightarrow a_{11}a_{22} - a_{12}a_{21}$

order 2  $\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow a_{11}(a_{22}a_{33} - a_{23}a_{32})$   
 $- (a_{12}(a_{21}a_{33} - a_{31}a_{23})$   
 $+ (a_{13}(a_{21}a_{32} - a_{31}a_{22}))$

Minors  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

for  $M_{11}$  (of  $a_{11}$ )  $= \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$

$M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$



(12)

Cofactor

$$O_{ij} \Rightarrow A_{ij} = (-1)^{i+j} M_{ij}$$

\* Adjoint

$$\text{adj}(A) = [A_{ij}]_{n \times n}^T$$

here  $A_{ij} \Rightarrow$  Cofactor matrix

Property  $(\text{adj } A) A = A (\text{adj } A) = |A| I_n$

Singular matrix

$$|A| = 0$$

Non singular matrix

$$|A| \neq 0$$

Inverse of a Matrix

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

 $|A| \neq 0$  means non-singularProperty

$$(AB)^{-1} = B^{-1} A^{-1} \quad |(A^T)^{-1}| = (A^{-1})^T$$

Rank of a Matrix $\rho(A)$  non-negative integer, (0 to  $\infty$ )\* at least one square sub-matrix of  $r$  order, whose determinant  $\neq 0$ 

Rank of Null matrix = 0

$$\text{Matrix } (m \times n) = \rho(A) \leq \min(m, n)$$

Echelon form

(i) a row which has all its elements zero below a row of non zero element

(ii) No. of zero before first non zero element in row  $<$  no. of zeros in next row.

Note  $P(A)$  in echelon = No. of non zero rows

for echelon form by elementary transformations  
make lower row = zero

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow \dots$$

Normal form

$$I_r \Rightarrow \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} Z_r \\ 0 \end{bmatrix}$$

$$P(A) = r$$

Normal form

after getting echelon form make identity matrix  
on constant by row & column

Nullity of a matrix for

$$AX = 0$$

rank nullity theorem:

here  $N_A$  = dimension of  
space (Nullity)

$$\text{rank} \leftarrow r_A + N_A = n \quad \begin{matrix} \downarrow \\ \text{nullity} \end{matrix} \quad \begin{matrix} \downarrow \\ \text{n.f. column} \end{matrix}$$

\* System of linear Simultaneous eq.

$$AX = B$$

$$AX = B$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Coeffi  
matrix

Vector  
(Solution)

Column

$B = 0$   
Homogeneous

$B \neq 0$

Inconsistent

$$P(A) \neq P(A:B)$$

$$B \neq 0$$

Trivial

Solution

$$P(A) = P(A:B) = n$$

(Unique)

$$B = 0$$

Non-trivial

Solution

$$P(A) = P(A:B)$$

$< n$   
( $\infty$  soln)

$$P(A:B)$$

$\downarrow$

Rank of augmented  
matrix



Solution methods

(i)

$$AX = B$$

$$\Rightarrow X = A^{-1}B$$

Invertible method

(ii)

Gauss elimination

A in echelon form by row transformation

Homogeneous system

$$AX = 0$$

 $|A| \neq 0 \Rightarrow \rho(A) = n \Rightarrow$  Trivial solution.all zero  $x_1 = x_2 = \dots = x_n = 0$  $|A| = 0 \Rightarrow \rho(A) = r < n$  is not trivial.at least one  $x \neq 0$ Linear Dependent  $\therefore$  Linear Independent

(LD)

(LI)

 $\rho(A) = \text{no. of variables}$  $\rho(A) \neq \text{no. of variables}$ Eigen values  $A = [a_{ij}]_{n \times n}$ Characteristic matrix  $[A - \lambda I]$   
of A $\lambda = |A - \lambda I|$  = characteristic polynomial  
of A $|A - \lambda I| = 0 \Rightarrow$  characteristic equationroots of equation = eigen values  
( $\lambda$ )• Sum of eigen value =  $\text{tr}(A)$ • Product =  $|A|$ • A &  $A^T$  same eigen value

• set of eigen value = spectrum of A

## Eigen vector (Character vector)

Put value of  $\lambda$  in equation

$$(A - \lambda I)x = 0$$

then find  $x$

here  $x$  is eigen vector of  $A$

① Diagonalization of matrix  $X_1, (x_1, x_2, \dots)$  Orthogonal

for

$$x_1^T x_2 = x_2^T x_1 = 0$$

$$A = P^{-1} D P$$

then  $D$

$P^{-1} A P$  is diagonal matrix  
the diagonal elements are  
eigen value of  $A$

## Caley - Hamilton Theorem

every square matrix satisfy its own  
characteristic equation

$$|A - \lambda I| = (-\lambda^n + c_1 \lambda^{n-1} + c_2 \lambda^{n-2} + \dots + c_n) = 0$$

$$\text{Put } A = \lambda$$

$$(-A^n + c_1 A^{n-1} + c_2 A^{n-2} + \dots + c_n I) = 0$$

↓ multiply by  $A^{-1}$  then divide by  $c_n$